Approximation of areas

1

a 117 **b** 6 **c** 2

2

a $\frac{1}{2}$ **b** $\frac{5}{16}$

3

a 60 **b** 17 000 **c** 34

4

a 2.426 **b** 1.167

5

a 180.828 **b** -21.000 **c** 0.919 **d** 0.502

e 1.165

6 5.2

7 1.5

8 256 units^2

9 $\frac{77}{60}$ units^2

10

a 62 units^2 **b** 50 units^2

11 16.1

12

a $y'' = 4x^2e^{x^2} + 2e^{x^2}$

b 30.5

c It is an overestimate as e^{x^2} is concave up between $x = 0$ and $x = 2$.

13 19.81 units^2

14 28.9

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Applications

15
a $232\,484\ \text{m}^2$ **b** $8183\ \text{m}^3$

16
a $14\,104\ \text{m}^2$ **b** $479.54\ \text{m}^3$

17 $432\ \text{m}^2$

18 $1\,707\,750\ \text{m}^2$

19 $26.8\ \text{m}^2$

20 $126\ \text{m}^2$

21 $3.12\ \text{ha}$

22 $201.95\ \text{m}^2$

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